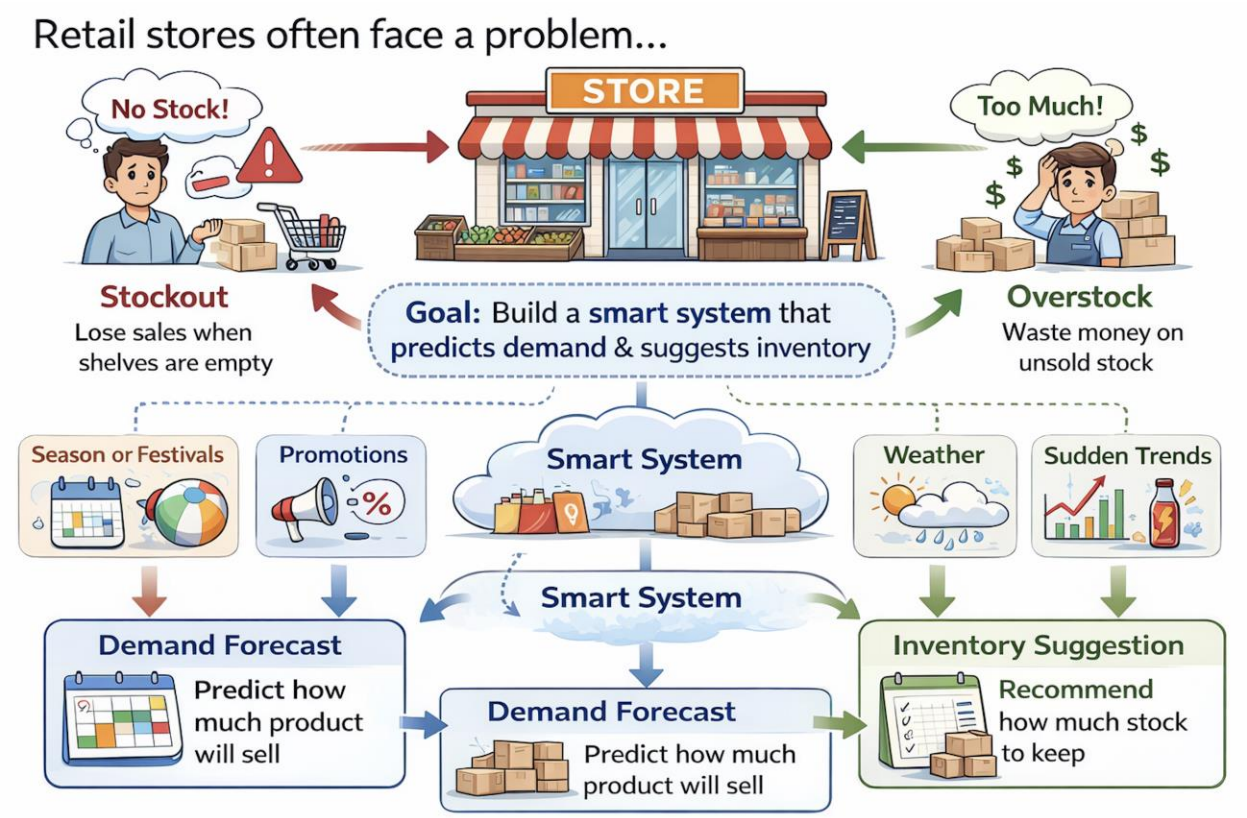
Smart Demand Forecasting for Retail Stores

AI for Intelligent Retail Operations

Retail businesses must ensure products are available when customers need them while avoiding excessive inventory. Predicting demand accurately is one of the most important and difficult problems in retail.

In this challenge, participants will build an **AI/ML system that forecasts product demand and recommends optimal inventory levels for retail stores.**



Problem Background

Retail stores frequently face two major operational challenges:

1. Stockouts

When a product is unavailable while customers want to buy it.

Impact:

- Lost revenue
- Poor customer experience
- Reduced customer loyalty

2. Overstocking

When stores keep excess inventory.

Impact:

- Increased storage costs
- Wasted capital
- Product expiry or markdown losses

These issues arise because **future demand is uncertain and influenced by many factors**, such as:

- Historical sales patterns
- Weekly and seasonal trends
- Promotions and discounts
- Weather conditions
- Festivals and holidays
- Local events
- Sudden demand spikes

Your task is to build an intelligent system that helps retailers **predict demand and maintain optimal inventory levels.**

Problem Statement

Design and build an **AI-powered demand forecasting system** that:

1. Predicts **daily product demand per store** for the next **7 days**

2. Recommends **optimal inventory levels** which suggests stockouts and overstocking
3. Presents insights through an **interactive dashboard** which shows:
    a. Forecast demand
    b. Historical sales trends
    c. Inventory recommendations
    d. Promotion and event effects
4. (Bonus) Adjusts forecasts using **real-time signals:** Add a **real-time adjustment mechanism** that updates demand predictions when sudden signals occur (e.g., promotions or weather changes)

Key Requirement

⚠️ **All data must be loaded into a database (SQL or NoSQL).**

Participants are free to choose any database.

What a Winning Solution Looks Like

- Accurate forecasts
- Practical inventory recommendations
- Clean data model using database
- Intuitive, business-friendly dashboard
- Clear explanation of logic

Dashboard Requirements

Your dashboard must clearly answer:

1. Demand Insights

- Predicted demand per product/store
- Trends over time
- Forecast vs historical comparison

2. Inventory Insights

- Recommended inventory levels
- Low stock alerts
- Overstock alerts

3. Business Insights

- Impact of promotions
- Impact of weather
- Event-driven demand spikes

4. Risk Indicators

- Stockout probability
- Demand volatility

Database Schema

Table Name	In Simple Words
sales_history	What sold, where, and when
products	What each product is
stores	Where each store is and what kind of store it is
promotions	Which products were on offer or discount
weather	Daily weather in each store's city
events_calendar	Festivals and events that may affect demand
realtime_signals	Sudden signals that may change demand immediately
test_input	The future records teams must predict
demand_forecast	The final prediction and inventory output table

Participants must create tables equivalent to the following schemas.

1. SALES_HISTORY

```
CREATE TABLE sales_history (
   date DATE,
   store_id INT,
   product_id INT,
   units_sold INT,
   sell_price FLOAT,
   inventory_on_hand INT,
   stockout_flag BOOLEAN,
   PRIMARY KEY (date, store_id, product_id)
);
```

2. PRODUCTS

```
CREATE TABLE products (
   product_id INT PRIMARY KEY,
   product_name VARCHAR(255),
   category VARCHAR(100),
   subcategory VARCHAR(100),
   brand VARCHAR(100),
   mrp FLOAT,
   shelf_life_days INT,
   lead_time_days INT
);
```

3. STORES

```
CREATE TABLE stores (
   store_id INT PRIMARY KEY,
   store_name VARCHAR(255),
   city VARCHAR(100),
   state VARCHAR(100),
   store_type VARCHAR(50),
```

```
  store_size_sqft INT
);
```

4. PROMOTIONS

```
CREATE TABLE promotions (
  date DATE,
  store_id INT,
  product_id INT,
  promo_type VARCHAR(50),
  discount_pct FLOAT,
  display_flag BOOLEAN,
  campaign_name VARCHAR(255)
);
```

5. WEATHER

```
CREATE TABLE weather (
  date DATE,
  city VARCHAR(100),
  avg_temp FLOAT,
  rainfall_mm FLOAT,
  humidity FLOAT,
  weather_condition VARCHAR(50)
);
```

6. EVENTS_CALENDAR

```
CREATE TABLE events_calendar (
  date DATE,
  event_name VARCHAR(255),
  event_type VARCHAR(100),
  city VARCHAR(100),
  impact_level VARCHAR(20)
);
```

7. REALTIME_SIGNALS (Bonus)

```
CREATE TABLE realtime_signals (
  timestamp TIMESTAMP,
  store_id INT,
  product_id INT,
  signal_type VARCHAR(50),
  signal_strength FLOAT,
  notes TEXT
);
```

8. TEST_INPUT

```
CREATE TABLE test_input (
  date DATE,
  store_id INT,
  product_id INT
);
```

9. DEMAND_FORECAST

```
CREATE TABLE demand_forecast (
  date DATE,
  store_id INT,
  product_id INT,
  predicted_units_sold FLOAT,
  recommended_inventory_level FLOAT
);
```

Data Relationships (Important)

- sales_history.store_id → stores.store_id

- sales_history.product_id → products.product_id
- promotions joins with sales_history
- weather.city joins with stores.city
- events_calendar.city joins with stores.city